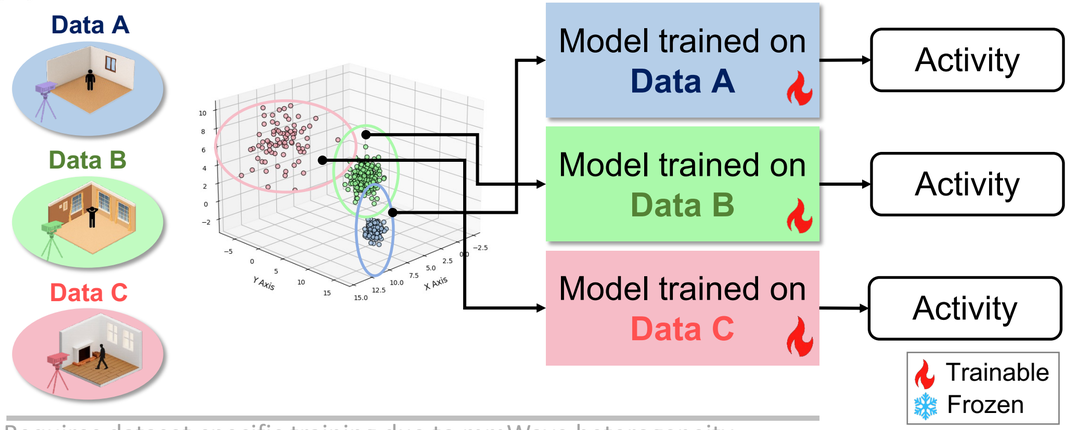
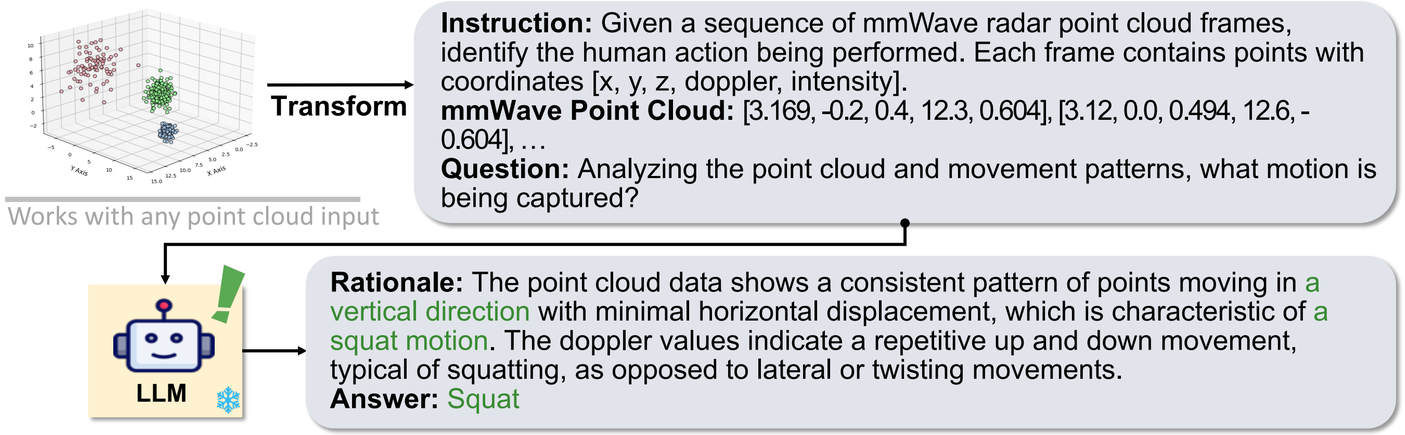
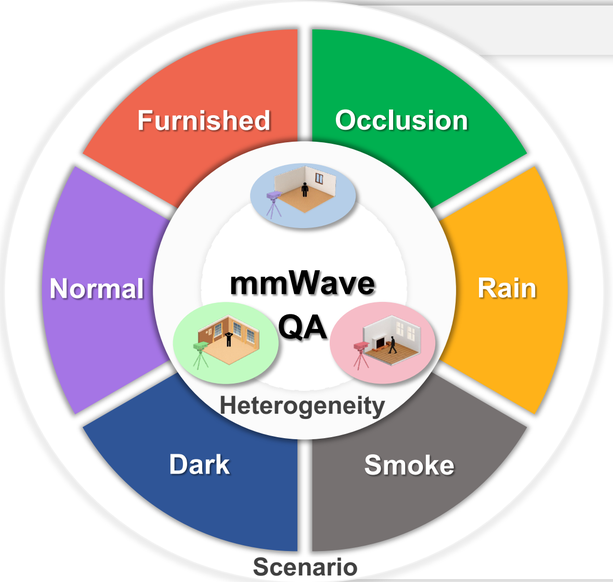
(a) Current mmWave Human Perception

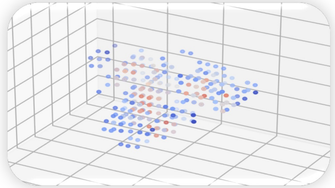


(b) LLM-integrated mmWave Human Perception (Training-free)



(c) Dataset Composition





mmWave Point Cloud

RGB Video (for reference)

1) ActRec: Action Recognition

Q: Which action is represented by this point cloud?
A: Basketball Dribble

2) TrajCheck: Spatial Movement Verification

Q: Has the person's position centroid changed location?
A: No

3) ActOrder: Temporal Action Sequencing

Q: What is the first action in this segment?
A: Basketball Dribble

4) ActNum: Action Cardinality Estimation

Q: How many distinct actions appear in this segment?
A: 1

5) LimbFocus: Primary Limb Motion Detection

Q: Which single body part is primarily moving in this action?
A: left arm